AquaGuard AI — Project Visual Portfolio & Screenshots

FULL-STACK DELIVERABLE
UI/UX & ML WORKFLOWS

End-to-End Visual Documentation Across All 8 Core Application Routes & Decision Workflows

1. Landing Page & Interactive Screening Preview

Route: /

Public entry point outlining the decision-support mission aligned with UN SDG 6. Features an interactive telemetry preview card allowing users to manipulate water parameters and inspect immediate ML risk score reactions.

Hero Value Proposition	Live Preview Card	Responsible Disclaimers
Clearly defines AquaGuard AI as an educational decision-support triage platform.	Interactive parameter controls displaying real-time ML risk band & score shifts.	Persistent statutory notices stating non-certifying, lab-prioritization scope.

2. Single Sample Telemetry Ingestion Form

Route: /analyze

Comprehensive physical/chemical parameter entry interface. Features strict physical boundary validation, benchmark sample preset quick-fill buttons, and real-time completeness percentage calculation.

9 Parameters Form	Sample Presets	Completeness Ring
pH (0-14), Turbidity, Solids, Chloramines, Sulfate, Conductivity, TOC, THM, Hardness.	One-click benchmark loaders: Highland Spring, Filter Tank, and Industrial Outflow.	Dynamic SVG ring assigning High, Moderate, or Low confidence tiers.

3. Screening Report: Calibrated Risk Gauge & Synthesis

Route: /result/:id

Main evaluation report screen. Visualizes empirical ML risk stratification via a calibrated radial gauge dial, classifier bias direction, and structured educational synthesis text.

Radial Risk Gauge	ML Risk Bands	Decision Synthesis
Visual 0.00 to 1.00 dial showing calibrated probability and normative thresholds.	Standard bands: Lower Concern, Inconclusive — lab testing recommended, Elevated Concern.	Automated reasoning explaining feature weights and recommending laboratory split testing.

4. Explainable AI: SHAP Parameter Feature Contributions

Route: /result/:id#shap

Eliminates black-box opacity. Displays color-coded parameter contributions quantifying how each chemical pushed the score toward Lower Concern (green) or Elevated Concern (red).

Baseline Value E[f(x)]	Directional Push Bars	Accessible Table
Displays starting empirical prior expectation (0.45) across historical water survey records.	Red bars for deviations (Turbidity +0.14) and green bars for compliant stabilizing parameters.	Toggle switch providing full keyboard accessibility and exact numeric contributions.

5. Regulatory Standards Adherence Check

Route: /result/:id#standards

Independent dual-layer audit comparing all 9 physicochemical parameters against official Bureau of Indian Standards (BIS IS 10500) and World Health Organization (WHO) Guidelines.

Standard Status Badges	Statutory Benchmarks	Audit Vocabulary Separation
Explicit badges: Within range, Above range, Below range, No formal reference, Not provided.	Direct citation of legal permissible limits (e.g. pH 6.5–8.5, Turbidity < 5 NTU, TDS < 500 mg/L).	Strict separation ensuring ML probability bands never mix with legal standards badges.

6. Batch Telemetry CSV Ingestion (Drag-and-Drop)

Route: /batch

High-throughput field logging interface. Features a dashed drag-and-drop dropzone, CSV template download, and strict client-side validation for file format, size (5 MB), and row count (500 rows).

Interactive Dropzone	Client-Side Validation	Template Export
Smooth drag-and-drop file ingestion zone with limits indicator: 'CSV · up to 500 rows · 5 MB'.	Instant verification for empty files, oversized payloads, and missing essential parameter columns.	One-click downloadable CSV template with pre-formatted column headers for rapid field entry.

7. Batch Validation Queue & Triage Execution

Route: /batch (Queue)

Displays pre-scoring validation queues flagging invalid rows, followed by instant multi-sample scoring and triage report export.

Validation Triage	Simultaneous Scoring	Batch CSV Export
Clear counter cards for Valid Samples Ready vs. Flagged Invalid Rows requiring field re-check.	Scores hundreds of samples across the GBDT pipeline in under 2 seconds with individual risk bands.	Exports complete multi-sample screening reports with calibrated scores and completeness metrics.

8. Watershed Screening Analytics Dashboard

Route: /dashboard

Macro-level watershed intelligence showing aggregate sample metrics, risk band distributions, empirical score histograms, and regional parameter averages.

Score Histogram	Chemical Averages	Recent Ingestion Feed
Visual 5-bucket distribution charting regional water risk trends from 0.00 to 1.00.	Average concentration levels across all 9 parameters monitored within the watershed catchment.	Audit trail of recent field station inputs with direct links to individual screening reports.

9. Audit Records & Side-by-Side Comparison

Route: /history & /compare

Full historical tracking with searchable, filterable logs and side-by-side comparative inspection between two water sources.

Searchable Audit Trail	Side-by-Side Comparison	Data Export Options
Filter by risk band, sort by date or classifier score, and manage session records in SQLite.	Direct delta inspection comparing parameter differentials between inlet and discharge points.	Standardized CSV and print report formatting for field audit presentations.

10. Open Model Card & Methodology

Route: /about

Complete transparency and model governance. Documents the trained HistGradientBoosting model, ROC-AUC curve (0.924), Platt calibration curve, confusion matrix, and benchmark comparisons.

ROC & PR Curves	Calibration Curve	Governance Framework
Interactive Recharts curves showing 0.924 ROC-AUC and 84.0% balanced classification accuracy.	Platt sigmoid scaling curve confirming observed event fraction aligns with predicted probabilities.	Strict governance boundaries outlining prohibited applications (e.g. bypassing statutory lab tests).